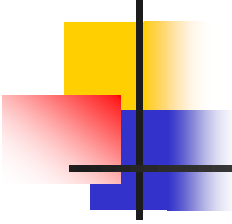


Part 1: Introduction

- Intelligence
- Defining A.I
- History
- Turing Test
- A.I Applications
- Multi-disciplinary domain



Intelligence

- Dictionary definition.
 - (1) : the ability to learn or understand or to deal with new or trying situations : **REASON**; *also* : the skilled use of reason
 - (2) : the ability to apply knowledge to manipulate one's environment or to think abstractly as measured by objective criteria (as tests)

-Merriam-Webster's Collegiate Dictionary

Intelligence – 2 views

- **Intelligence = Autonomous movement**
- Intelligence = Thinking



Vaucanson
18th century



Shakey (1970)

Stanford Research Institute



SONY Aibo (1998)

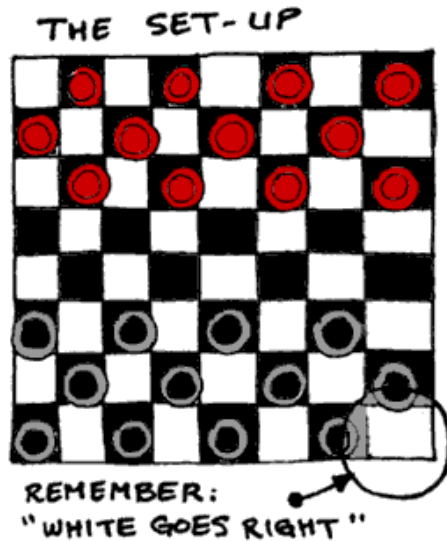
Intelligence – 2 views

- Intelligence = Autonomous movement
- **Intelligence = Thinking**



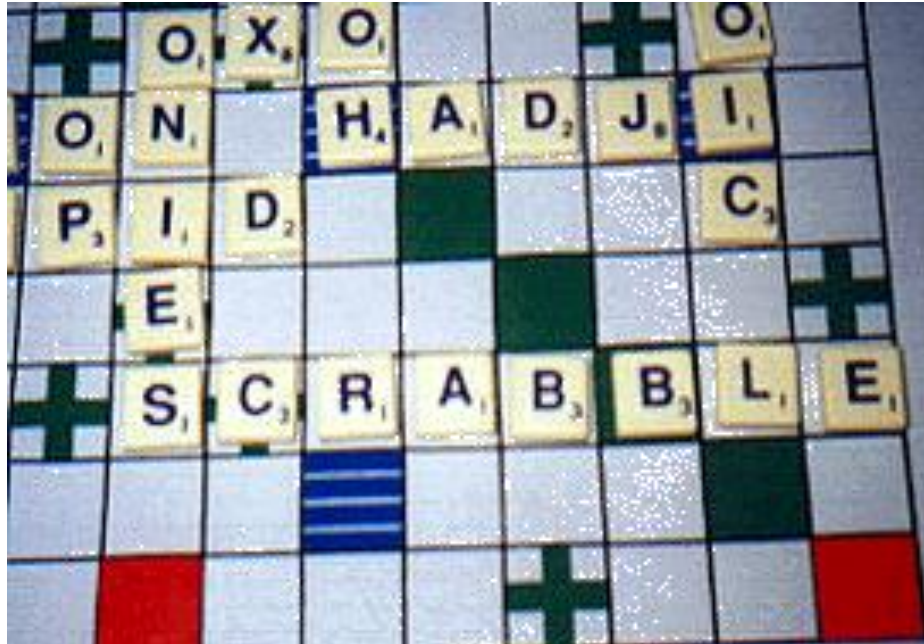
Deep Blue defeats Garry Kasparov (1997)

Playing games:Draughts



Is a computer more intelligent
if it beats you in a game of draughts?

Playing games: Scrabble

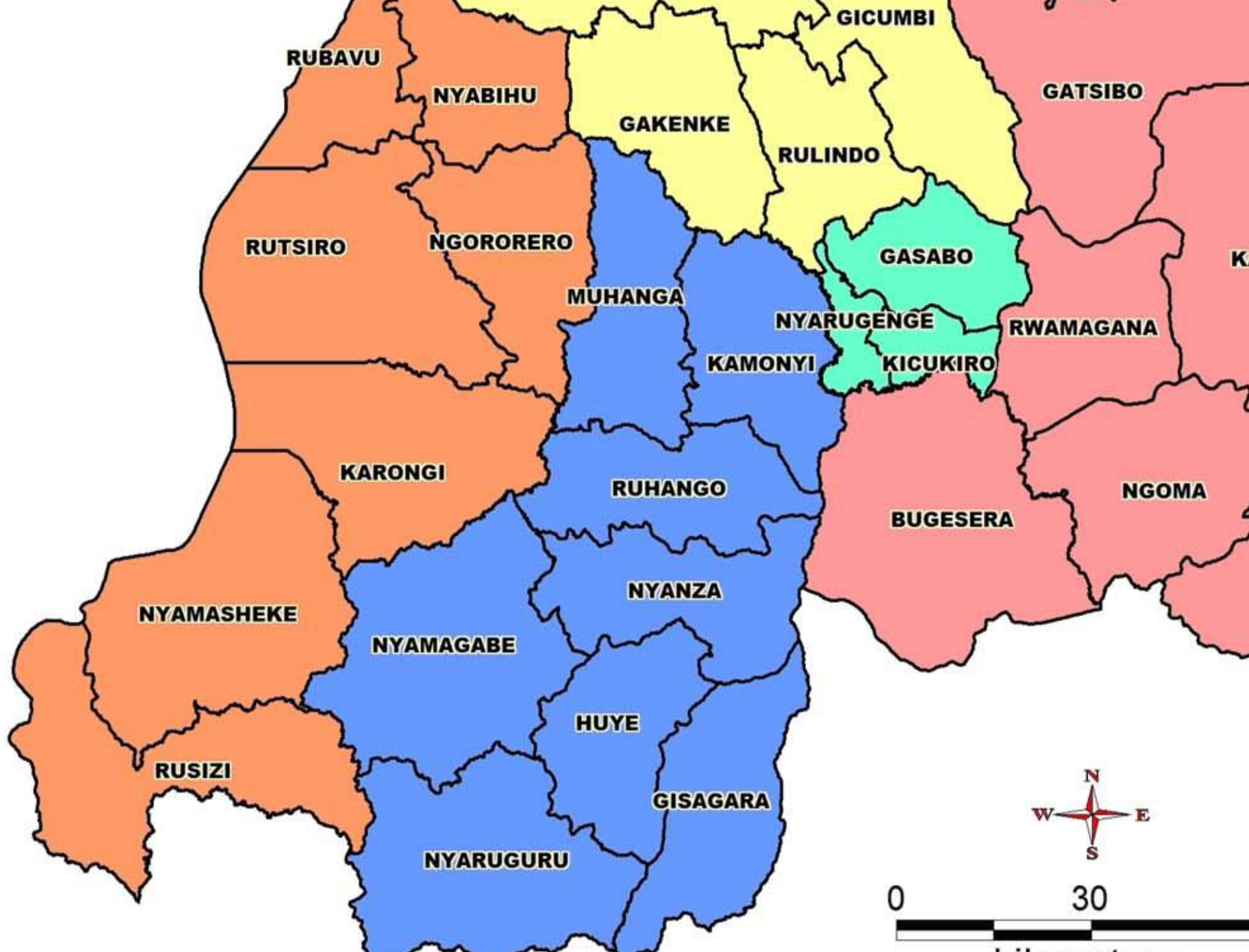


Is a computer more intelligent
if it beats you in a game of scrabble?

Playing games: Jigsaw



Is a computer more intelligent if it solves the puzzle faster than you do?

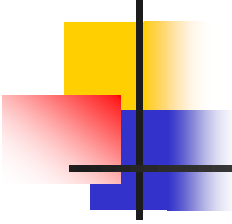




Defining A.I

There is no agreed definition of the term artificial intelligence. However, there are various definitions that have been proposed. These are considered below.

- AI is a study in which computer systems are made that think like human beings. Haugeland, 1985 & Bellman, 1978.
- AI is a study in which computer systems are made that act like people. AI is the art of creating computers that perform functions that require intelligence when performed by people. Kurzweil, 1990.
- AI is the study of how to make computers do things which at the moment people are better at. Rich & Knight
- AI is a study in which computers that rationally think are made. Charniac & McDermott, 1985.
- AI is the study of computations that make it possible to perceive, reason and act. Winston, 1992



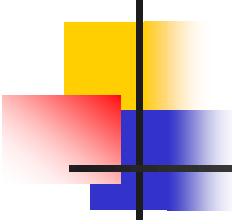
Defining A.I

- AI is the study in which systems that rationally act are made. AI is considered to be a study that seeks to explain and emulate intelligent behaviour in terms of computational processes. Schalkeoff, 1990.
- AI is considered to be a branch of computer science that is concerned with the automation of intelligent behavior. Luger & Stubblefield, 1993.
- ...



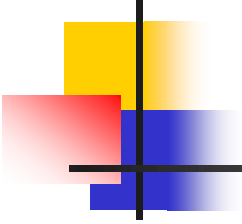
Goals of AI

- The definitions of AI gives four possible goals to pursue:
 1. Systems that think like humans
 2. Systems that think rationally
 3. Systems that act like humans
 4. Systems that act rationally



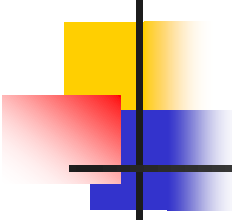
General AI Goals

- Replicate human intelligence
- Solve knowledge intensive tasks
- Make an intelligent connection between perception and action
- Enhance human-human, human-computer, computer to computer interaction/communication



..but the Objectives of A.I

- Understand how living beings behave, think, learn.
- Engage in experiments by building artificial systems
- Derive applications for robotics and computers



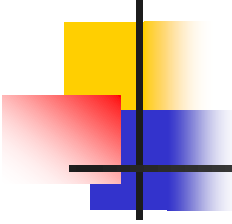
AI Approaches

- Cognitive science: think like human
- Laws of thought: Think rationally



Think human-like

- An exciting new effort to make **computers think**; that it is, the machines with minds, in the full and literal sense.
- Focus is not just on behavior and I/O, but looks at **reasoning process**.
- Computational model as to **how** results were obtained.
- **Goal** is not just to produce **human-like behavior** but to produce a sequence of steps of the reasoning process, similar to the steps followed by a human in solving the same task.



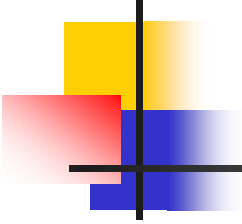
Think rationally

- The study of mental faculties through the use of computational models; that it is, the study of the computations that make it possible to *perceive, reason, and act*.
- Focus is on *inference* mechanisms that are provably correct and guarantee an optimal solution.
- Develop systems of representation to allow inferences to be like *"Socrates is a man. All men are mortal. Therefore Socrates is mortal."*
- *Goal* is to formalize the *reasoning process* as a system of logical rules and procedures for inference.
- The issue is, not all problems can be solved just by reasoning and inferences.



Turing test

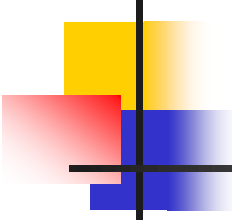
- The art of creating machines that perform functions requiring **intelligence** when performed by people; that it is the study of, how to make computers do things which at the moment people do better.
- Focus is on action, and not **intelligent behavior** centered around representation of the world.
- A Behaviorist approach, is not concerned with how to get results but to the similarity to what human results are.

- 
-
- Goal is to develop systems that are human-like



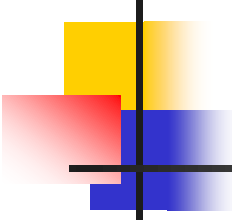
Rational Agent: Act rationally

- Tries to explain and emulate **intelligent behavior** in terms of computational processes; that it is concerned with the automation of intelligence.
- Focus is on systems that act sufficiently if not optimally in all situations;
- It is passable to have imperfect reasoning if the job gets done.
- **Goal** is to develop systems that are **rational** and sufficient.



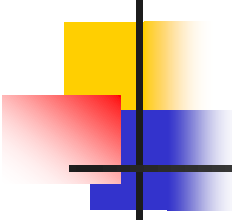
AI Techniques

- Various techniques that have evolved, can be applied to a variety of AI tasks. The techniques are concerned with how we represent ,manipulate and reason with knowledge in order to solve problems.



Examples

- Describe and match
- Goal reduction
- Constraint satisfaction
- Tree searching
- Generate and test
- Rule based systems
- Neural network
- Reinforcement learning
- Genetic algorithm

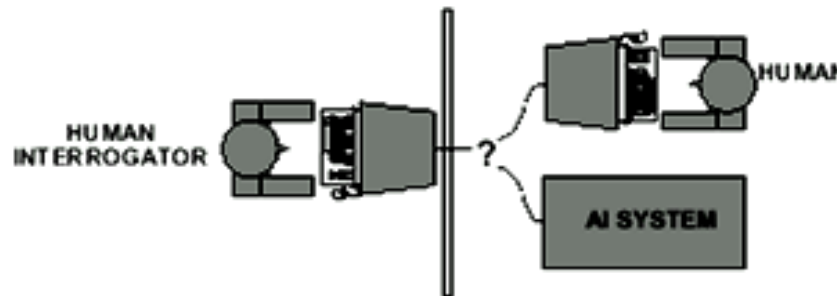


History

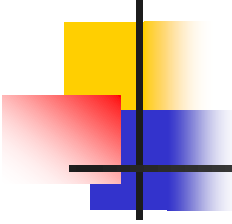
- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1950s Early AI programs, including Samuel's checkers (draughts) program
- Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1966–74 AI discovers computational complexity, Neural network research almost disappears
- 1969–79 Early development of knowledge-based systems
- 1980–88 Expert systems industry booms
- 1988–93 Expert systems industry busts: "AI Winter"
- 1985–95 Neural networks return to popularity
- 1988– Resurgence of probabilistic and decision-theoretic methods
Rapid increase in technical depth of mainstream AI,
"Nouvelle AI": ALife, GAs, soft computing

Acting Humanly: Turing Test

- Alan Turing's 1950 article *Computing Machinery and Intelligence* discussed conditions for considering a machine to be intelligent
 - “Can machines think?” $\longleftrightarrow$ “Can machines behave intelligently?”
 - The Turing test (The Imitation Game): Operational definition of intelligence.



- **Computer needs to posses:** Natural language processing, Knowledge representation, Automated reasoning, and Machine learning
- **Problem:** 1) Turing test is not reproducible, constructive, and amenable to mathematic analysis. 2) What about physical interaction with interrogator and environment?
- **Total Turing Test:** Requires physical interaction and needs perception and actuation.



A.I applications

- ALVINN
- MYCIN, PROSPECTOR,...
- Deep Blue
- Data Mining
- ROBOTICS
- ...

ALVINN

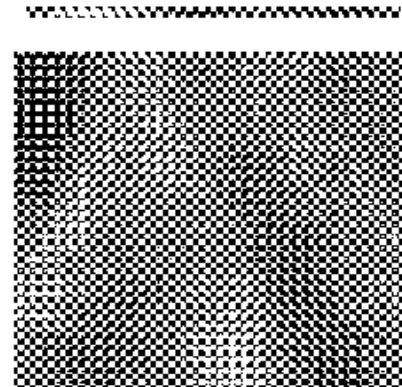
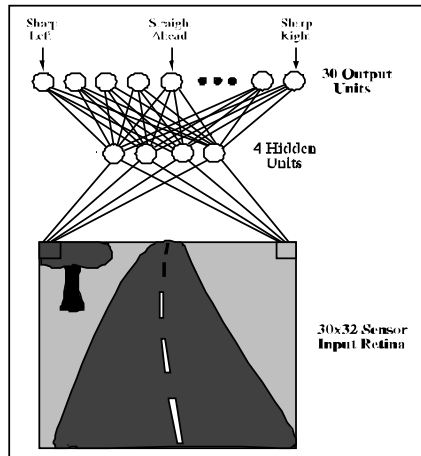
Drives 70 mph on a public highway



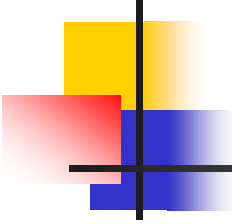
30 outputs
for steering

4 hidden
units

30x32 pixels
as inputs



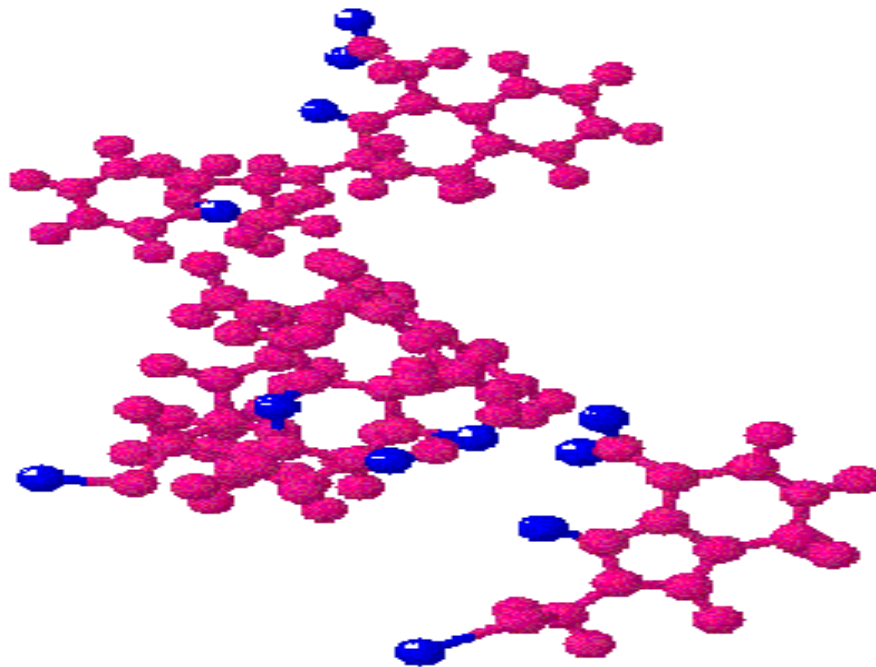
30x32 weights
into one out of
four hidden
unit



Knowledge-based systems

- Medical Diagnosis - MYCIN
 - 1971, A program that could diagnose blood infections. It had 450 rules
- Mineral Prospecting - PROSPECTOR
 - 1979, A program that with geological data. It recommended exploratory drilling sites that proved to have substantial molybdenum deposits.

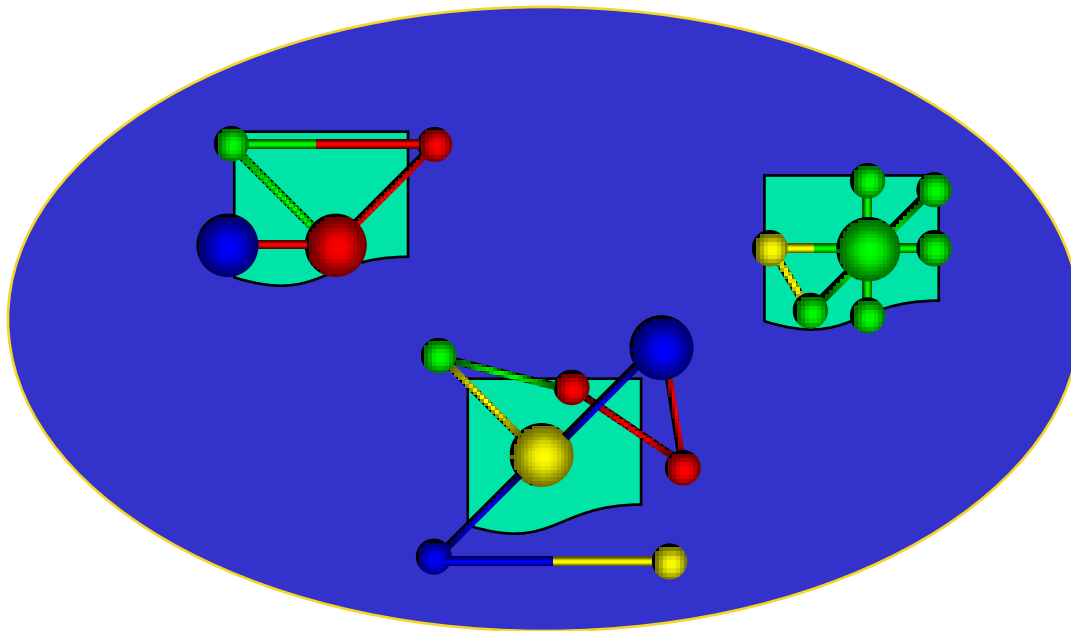
Data Mining



structure of new molecules indicate that they may cause cancer ?

Data Mining

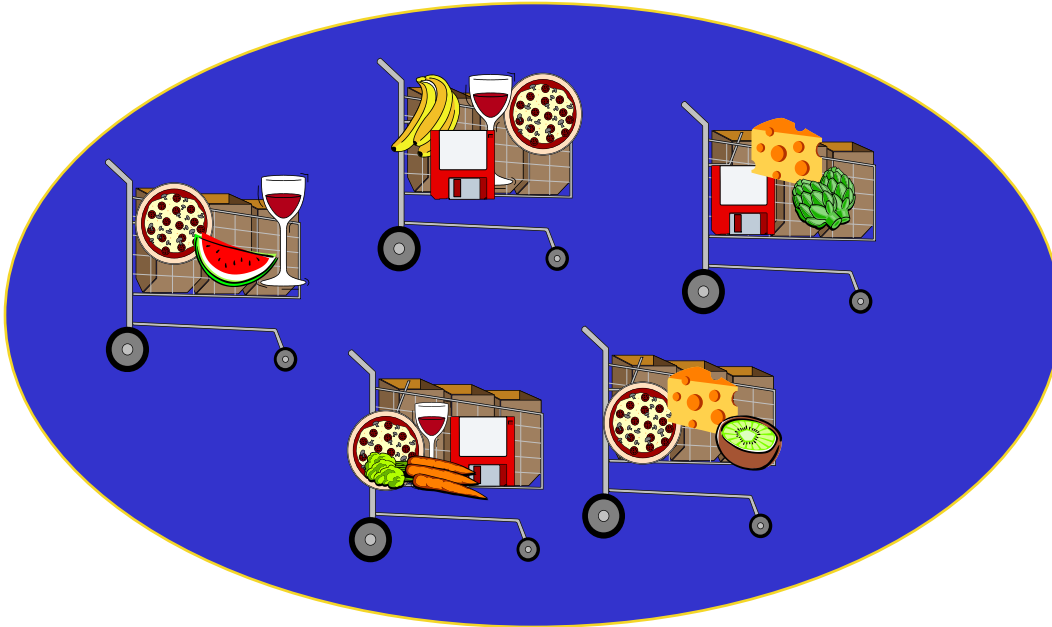
- An application of Machine Learning techniques
 - It solves problems that humans can not solve, because the data involved is too large, noisy ...



Detecting cancer risk molecules is one example.

Data Mining

- A similar application:
 - In marketing products ...

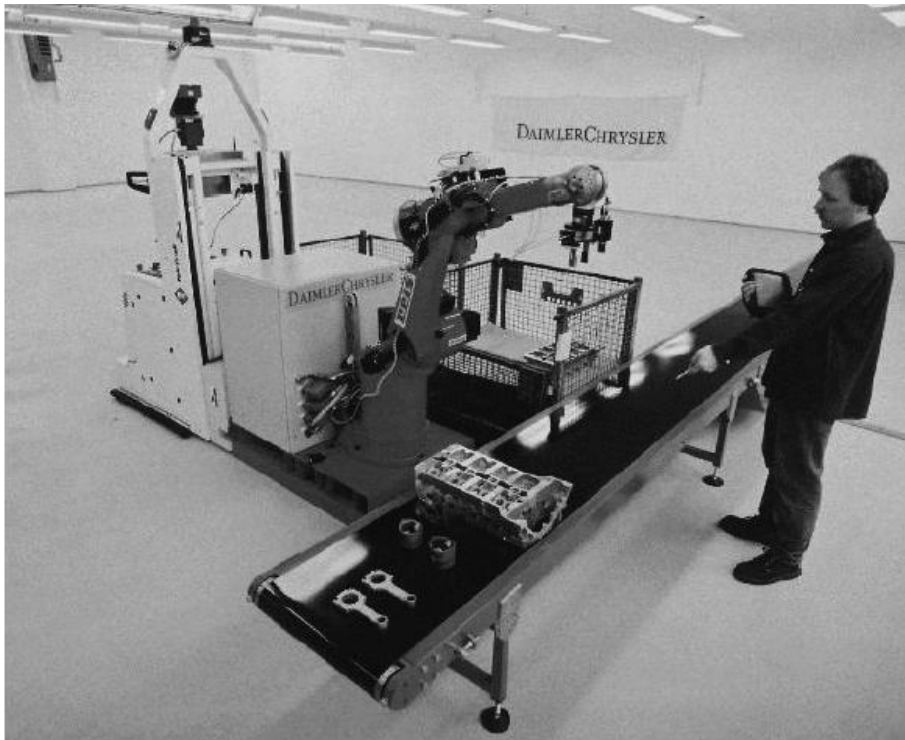


Predicting customer behavior in supermarkets is another.

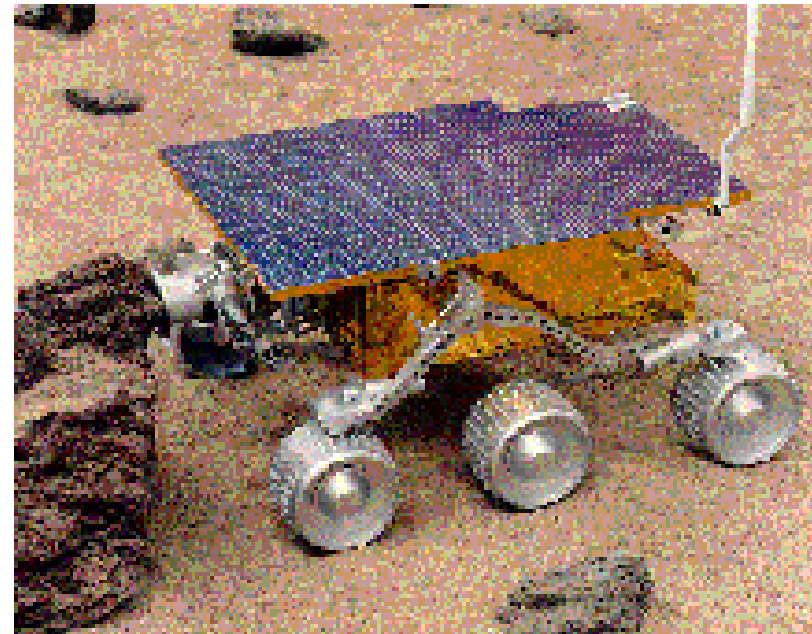
- Fraud Detection
 - 1997, Credit card fraud detection

Robotics

- Traditional Robots



- Mars Rover (1996)
 - Exploring Mars

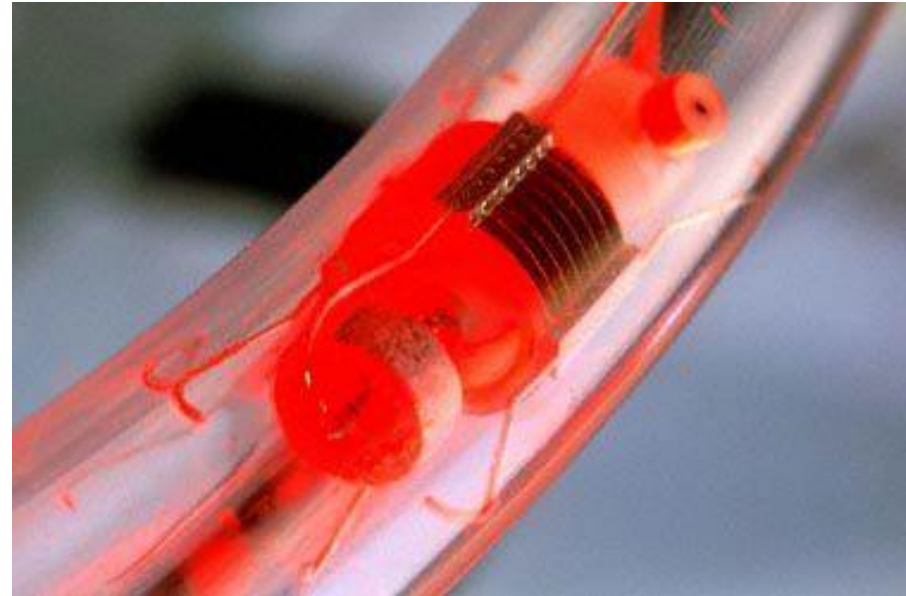


Robotics

- Wire repair



- Pipe inspector





Branches of A.I

- Machine vision
- Speech synthesis and recognition
- Machine Learning
- Robotics
- Natural Language and understanding
- Problem solving
- Game playing
- ..

Applications

- Computer vision
- Image Recognition
- Language and speech processing





AI, a Multi-disciplinary domain

- Engineering:
 - robotics, vision, control-expert systems, biometrics,
- Computer Science:
 - AI-languages , knowledge representation, algorithms, ...
- Pure Sciences:
 - statistics approaches, neural networks, fuzzy logic, ...
- Linguistics:
 - computational linguistics, phonetics and speech, translation systems, ...
- Psychology:
 - cognitive models, knowledge-extraction from experts, ...
- Medicine:
 - human neural models, neuro-science,...



How far is the field advanced?

- Very far
 - Humanoid robots
 - Natural language interfaces
 - Expert systems
 - ...
- Not far at all compared to humans
 - Integration?
 - Learning?